YU LIU

yu130dfb4@gmail.com | aquareo.github.io/en/

Ph.D. applicant in Applied Mathematics / Scientific Computing — Fall 2027

RESEARCH INTERESTS

Numerical analysis; scientific computing; stochastic analysis.

EDUCATION

The Chinese University of Hong Kong, Shenzhen 2022–2024

M.Phil. in Mathematics | Cumulative GPA: 3.436/4.000.

Thesis: *Pricing and Trading Strategies of Chinese Convertible Bonds.* Advisor: Prof. Gongqiu Zhang.

Tongji University 2016–2020

B.Sc. in Mathematics and Applied Mathematics.

Major-course average: 85.09/100; overall equivalent average: 82.11/100 (official transcript).

Thesis: *Image Restoration: a Wavelet Frame Based Approach and Beyond.* Advisor: Prof. Jae Kyu Choi.

Selected A-grade coursework: Mathematical Analysis I–III, Real Analysis, Theory of Probability, Numerical Analysis (3 credits), Computational Method.

RESEARCH EXPERIENCE

Numerical Valuation and Multi-Factor Strategies for Chinese Convertible Bonds 2023–2024

M.Phil. research, CUHK-Shenzhen

- Developed a Monte Carlo valuation framework with path-dependent contractual provisions, least-squares continuation-value estimation, and backward dynamic programming; compared global and stock-price-region regressions.
- Compared model and market prices for ten AAA-rated convertible bonds and evaluated an undervaluation-based selection strategy against a double-low benchmark.
- Evaluated factors using IC analysis and stratified backtesting; compared linear and polynomial regression, SVM, random forests, XGBoost, LightGBM, and MLP with 7-, 10-, and 20-day rolling training windows.
- Constructed SVM and linear-regression multi-factor strategies and evaluated them in a 118-trading-day simulation with a 0.1% transaction-cost assumption. The thesis covers both pricing and multi-factor strategies; the related preprint focuses on pricing.

Image Restoration and Numerical Optimization with Wavelet Frames

Undergraduate thesis, Tongji University

- Studied analysis-based, synthesis-based, and balanced wavelet-frame reconstruction models and their connections to sparse regularization and soft thresholding.
- Worked through ADMM / Split Bregman formulations for the analysis-based model, FFT-based deblurring subproblems, and assumptions behind existing convergence results.
- Compared B-spline and Haar frame methods with total variation for deblurring and inpainting, evaluating PSNR, HFEN, SSIM, iteration counts, and runtime.

PREPRINT

Liu, Y. *Valuation Model of Chinese Convertible Bonds Based on Monte Carlo Simulation.* arXiv:2409.06496 (2024; updated 2026). Preprint; not peer-reviewed.

TEACHING EXPERIENCE

Teaching Assistant, MAT1012 Honours Calculus II 2024

The Chinese University of Hong Kong, Shenzhen

- Prepared and taught tutorials and exam-review sessions covering proof-based and computational problems; graded assignments, quizzes, and examinations; supported individual problem-solving discussions.

PROFESSIONAL EXPERIENCE

Backend Developer | Mavericksmobile, Chengdu Jan–Apr 2026

Contributed to AI application backends, task-processing workflows, monitoring, and maintenance.

Backend Software Developer | BYD, Shenzhen Sep 2024–Jul 2025

Developed backend services, content-management features, and workflow automation.

PROGRAMMING AND LANGUAGES

C++, Python, MATLAB, Java. English: IELTS 7.0.